SAM-RSIS: Progressively Adapting SAM With Box Prompting to Remote Sensing Image Instance Segmentation

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***Abstract*— The recent segment anything model (SAM) trained on massive close-range images has demonstrated impressive performance on general segmentation or specific segmentation tasks with manual prompts. However, the significant domain shift problem between remote sensing and close-range images should be tackled before introducing the pretrained SAM to remote sensing instance segmentation (RSIS). To address this and unlock the potential of SAM in RSIS, this article proposes a novel framework called SAM for remote sensing instance segmentation (SAM-RSIS), which overcomes the problems in a few recent works that only adapt a part of SAM to remote sensing. SAM-RSIS fine-tunes the vision transformer (ViT) backbone and mask decoder of SAM progressively on remote sensing data and uses automatic box prompting to eliminate the need for manual prompting. SAM-RSIS consists of an object detection stage and a mask generation stage. In object detection, we introduce an adapter to adapt knowledge embedded in the pretrained ViT backbone to remote sensing images and then build an object detector. In mask generation, using the detected bounding boxes as prompts, along with two learnable mask output tokens, and the two-layer high-resolution features from the adapter, we fine-tune the mask decoder of SAM to produce high-quality masks. Experimental results on the WHU, WHU-Mix, and NWPU datasets for binary and multiclass RSIS demonstrate the effectiveness and robustness of the proposed method, surpassing various derivative methods of SAM and achieving performance comparable to and even better than the specific state-of-the-art instance segmentation methods.**

***Index Terms*— Efficient tuning, foundation model, instance segmentation, remote sensing, segment anything model (SAM).**

I. Introduction

AUTOMATED and intelligent means for the extraction and monitoring of key elements of the land surface (e.g., buildings, ports, and vehicles) from high-resolution remote sensing images hold significant importance for urban planning, natural disaster management, and other fields [1], [2]. The technology called instance segmentation can recognize individual objects in an image and perform pixel-level classification to obtain their fine-grained boundaries. This urgently calls for the development of instance segmentation methods for interested object recognition on high-resolution remote sensing imagery. In this article, we propose an instance segmentation method based on the segmentation foundation model, segment anything model (SAM), to boost the instance segmentation performance on remote sensing images.

Researchers have recently made significant efforts to advance close-range instance segmentation performance in the field of computer vision [3], [4]; those modern deep learning-based methods have far surpassed traditional methods [5], [6]. This also happened in remote sensing [7], [8]. A modern instance segmentation network for remote sensing imagery usually adopts a convolutional neural network (CNN) backbone (e.g., ResNet series [9]) or vision transformer (ViT) backbone [10] pretrained on the ImageNet dataset [11], [12], which is then retrained or fine-tuned, along with the instance segmentation decoder, on remote sensing data. More recently, due to contrastive learning [13], [14], masked image modeling [15], cyclic annotation [16], and other techniques, emerging foundational models, such as SAM [16], DINO v2 [17], SegGPT [18], and so on, can be trained on even larger scale datasets than ImageNet. Therefore, they exhibit stronger feature abstraction and generalization capabilities than common pretrained backbone models, such as ResNet 50 [9]. Among them, SAM, utilizing ViT [10] as the backbone, has designed a generic image segmentation framework pretrained on 1 billion masks and 11 million images of the SA-1B dataset [16]. SAM can perform zero-shot segmentation or generate masks for the objects of interest based on manual prompts, such as bounding boxes or points, showing impressive generalization capabilities on natural scene images. Observing the promising performance of SAM on natural scene images, we aim to explore its potential for instance segmentation on remote sensing images. However, SAM has the following issues that prevent us from directly applying it to instance segmentation in remote sensing images.

First, there is a significant domain gap between top-view remote sensing images and close-range natural scene images. Since the capability of deep learning models is restricted to the distribution coverage of their training data [19], it is not surprising that directly applying the original SAM trained

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on natural scene images to remote sensing images is highly challenging [20], [21].

Second, SAM addresses a general, category-agnostic segmentation problem. That is to say, SAM learns features from geometric shapes and structures, and it may have learned semantic information but not directly. Consequently, it may struggle to understand complex geographical and semantic context information in remote sensing images. Additionally, manual prompts are required to guide SAM with semantic category information, which is time-consuming and labor-intensive, thereby limiting its application.

Third, SAM faces challenges in generating complete segmentation results for complex or small-sized objects in low-resolution images [20]. In SAM, the input image size for the ViT backbone is fixed at 1024×1024 pixels, and the output feature size is 1/16 of the input, i.e., 64×64 pixels. That is to say, the mask decoder must restore a segmentation map of full resolution from features of size 64×64. Therefore, objects with small sizes and blurred boundaries in a remote sensing image may not be recovered well from low-resolution feature maps.

There have been a few methods [21], [22], [23] that have tried to adapt SAM from natural scene images to remote sensing images. These methods either directly use the output of SAM as an independent auxiliary for downstream tasks (Ma et al. [21]) or only fine-tune or modify a small part of SAM's structure, such as the mask decoder (GeoSAM [22]) or the prompt encoder (RSPrompter [23]), which limit SAM's ability to address the significant domain shift between natural scene images and remote sensing images. More comprehensive and deeper explorations of different parts of the SAM pipeline are required for a more successful utilization on RSIS while retaining SAM's strong generalization ability.

This article proposes an adaptation method called SAM for remote sensing instance segmentation (SAM-RSIS), which addresses the aforementioned three issues and enables the adaptation of SAM for instance segmentation on remote sensing images. Unlike previous studies that only modify mask decoder [22] or prompt encoder [23], our SAM-RSIS concerns how to extract better trainable features from the ViT backbone, how to feed the prompt encoder, and how to adjust the mask decoder in remote sensing images more comprehensively. SAM-RSIS has two stages. In the first stage, we fine-tune SAM's ViT backbone using a multiscale adapter and simultaneously train a new object detector. The adapter handles the domain gap problem between natural scene images and remote sensing images, and the object detector automatically obtains object categories and generates bounding box prompts for the second stage. In the second stage, the generated boxes are input into the prompt encoder of SAM to fine-tune the mask decoder. High-resolution feature's output by the multiscale adapter is also introduced to provide richer semantic information. Two additional learnable output tokens are designed for predicting masks at different scales. The main contributions of this article can be summarized as follows.

1) We develop a two-stage adaptation method named SAM-RSIS, which progressively adapts both SAM's ViT backbone and mask decoder to remote sensing images and performs instance segmentation via automatic box prompting.
2) A multiscale adapter is designed to alleviate the domain shift problem and provide high-resolution features for remote sensing objects, an object detector is introduced to address the category-agnostic issue in SAM and providing box prompts, and two output tokens are newly added for the multiscale problem in remote sensing.
3) We discover that: first, our SAM-RSIS surpasses all the other similar derivative methods of SAM; second, it has not surpassed those specific SOTA methods for RSIS in an "interpolation" case, i.e., training and testing sets are from similar scenes; third, but most importantly, our method shows the best performance and superior generalization ability in an "extrapolation" case, i.e., training and test data separately captured from different cities around the world.

II. Related Work

A. Instance Segmentation

As most instance segmentation methods are derived from object detection methods, the existing instance segmentation methods can be categorized following the division criteria of object detection methods, namely, based on the number of stages and the requirement for anchors. The two-stage anchor-based instance segmentation methods mainly follow the object detection pipeline of faster R-CNN [24], which includes region proposal generation as the first stage and box regression and classification the second stage. Mask R-CNN [5] adds a mask head to faster R-CNN for handling the instance segmentation task. Many researchers have built their new ideas upon mask R-CNN [25], [26]. RetinaNet [27] is a one-stage anchor-based object detection method that directly predicts boxes using the features of the feature pyramid network (FPN) and addresses the imbalance of positive and negative samples with the focal loss. YOLACT [6] follows RetinaNet and predicts instance masks by linearly combining the prototype generated by the prototype mask prediction head and the mask coefficients from the box head.

Unlike dense anchor-based methods, anchor-free methods [28], [29] directly predict object locations, eliminating the need for positive and negative sample matching and providing greater flexibility. SOLO v1 [30] and v2 [31] are one-stage anchor-free instance segmentation methods inspired by YOLO [32], and they transform the instance segmentation problem into a position classification problem and classify and predict masks for objects on the divided grid. CondInst [33] follows the framework of one-stage anchor-free object detection network FCOS [28] and introduces relative position learning to better distinguish the semantic features of two objects belong to the same class. Recent MaskFormer [34] and Mask2Former [35] are the popular transformer frameworks, which can solve instance, semantic, and panoptic segmentation tasks together.

Due to the distinct characteristics of objects in remote sensing images compared to natural scene images, recent years have seen researchers exploring specialized methods for

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remote sensing instance segmentation (RSIS). For single-class RSIS, Wei et al. [36] predict building pixels by performing semantic segmentation and then transform segmentation maps into individual building polygons by a regularization algorithm. BuildMapper [37] extracts building instances through learnable contour evolution. Mou and Zhu [38] address vehicle instance detection by integrating semantic segmentation and boundary detection. For multiclass instance segmentation, some methods aim to enhance the ability of the instance segmentation network to handle multiscale objects. For example, Zhang et al. [39] propose a one-stage network solving the problem of undersegmentation of geospatial instances with large varying scales. Chen et al. [40] introduce a dense FPN to learn the hierarchical detailed features of objects. Additionally, some approaches focus on accurately segmenting object boundaries. Gong et al. [41] introduce a penalty map in mask R-CNN to better distinguish the edges of different objects. Furthermore, some methods enhance the performance of instance segmentation models from multiple perspectives. For example, Ye et al. [42] design three modules—namely, the channel-spatial attention module, the multiscale aware module, and the semantic relation learning module—to enhance the feature representation capability of mask R-CNN. Similarly, the study by Yang et al. [43], also built upon mask R-CNN, uses adaptive and dynamic feature learning modules to simultaneously address the challenges of object size variation and instance boundary localization.

To date, deep learning-based instance segmentation methods still struggle with poor generalization ability in practical applications. Adapting and transferring foundation models pretrained on extremely large datasets to RSIS offer a new opportunity.

B. Segment Anything Model

The concept of a foundation model originates from the field of natural language processing (NLP) and is defined as a model trained on large-scale data in a self-supervised or semi-supervised manner that can be adapted for downstream tasks [44]. Seeing the significant advancements of foundation models in the domain of NLP, research on the visual foundation model (VFM) has also become a hot topic [15], [17]. The SAM is the first VFM specifically designed for image segmentation. SAM is a large promptable model that can return a valid segmentation mask given any type of prompts (box, point, etc.) inspired by the prompting learning technique proposed in the field of NLP [45], [46]. As shown in Fig. 1, SAM can be divided into three parts, i.e., image encoder, prompt encoder, and mask decoder. The image encoder is a ViT backbone used for image feature extraction, which takes the images of 1024×1024 pixels as inputs and outputs image embeddings of 64×64 pixels. The prompt encoder encodes the position information from the input prompts into embedding representations. The mask decoder consists of two attention layers, each consisting of a prompt self-attention block and bidirectional cross-attention blocks (prompt-to-image attention and vice versa). The image embeddings, the concatenated position embeddings, and learnable output tokens (three mask tokens and an IoU token) are inputted into the attention layers for updating. Then, the updated image embeddings are upsampled from 64×64 pixels to 256×256 pixels and transformed into three prediction masks using the updated mask tokens, and each mask has its corresponding IoU score. Finally, the mask with the largest score is used as the final mask. SAM is trained on a very large close-range dataset SA-1B, containing over 1 billion masks and 11 million images, showing its great zero-shot generalization ability on natural scene images. However, SAM cannot be straightly applied to the RSIS task, fine-tuning, and adaptation are required.

C. Adapting SAM to Downstream Tasks

The original SAM paper [16] reports that SAM with the ViT-Huge backbone was trained for 68 h with 256 A100 GPUs. Therefore, it is very expensive to retrain the whole SAM pipeline for downstream tasks. A simple idea is to freeze the pretrained ViT backbone of SAM and replace the backbone model (e.g., Resnet 50) of a task network with it, training only the remaining part of the network. However, this cannot fully leverage the complete SAM architecture.

Fine-tuning is the mainstream way for adapting foundation models to new applications. Currently, the fine-tuning methods primarily fall into the categories of adapter-based [47], prompt-based [48], and reparameterization-based [49]. Adapter-based methods insert lightweight adapters into a foundation model and only tune the parameters in adapters to adapt to downstream tasks while keeping the pretrained backbone frozen [50]. SAM-Adapter [47] incorporates simple adapters into SAM and shows performance gains in camouflaged object detection and shadow detection tasks. Subsequently, SAMUS [51] and 3DSAM-adapter [19] design different adapters to make SAM better adapt to the medical image domain. Prompt-based methods [48] focus on designing or changing prompts to adjust the model to different downstream tasks. Reparameterization-based methods, such as LoRA [52] and its variants Q-LoRA [53], optimize the low-rank matrices to approximately update model parameters. Following this, SAMed [49] applies LoRA to adapt SAM to the medical image segmentation.

Zero-shot or few-shot learning strategies are another way for utilizing SAM in downstream tasks. Grounded-SAM [54] combines the open-world object detector grounding-DINO [55] and SAM. The class names provided as text prompts and the image are inputted into the pretrained grounding-DINO to output bounding boxes, which are then fed into the pretrained SAM to generate corresponding masks. Zhang et al. [56] propose a training-free one-shot strategy named PerSAM and its fine-tuning variant, PerSAM-F. The method uses one or some reference masks on an image as a guide for SAM for segmenting other object masks similar to the reference. Obviously, this type of approach requires similar data distribution between the segmented image and the training datasets of SAM, limiting its practical application.

A few studies have attempted to apply SAM to the field of remote sensing. Ma et al. [21] use the output of SAM to assist

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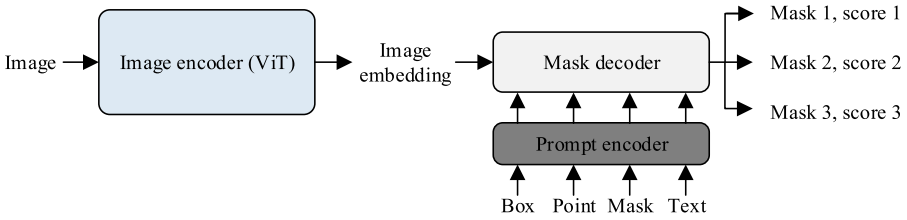


Fig. 1. Structure of SAM.

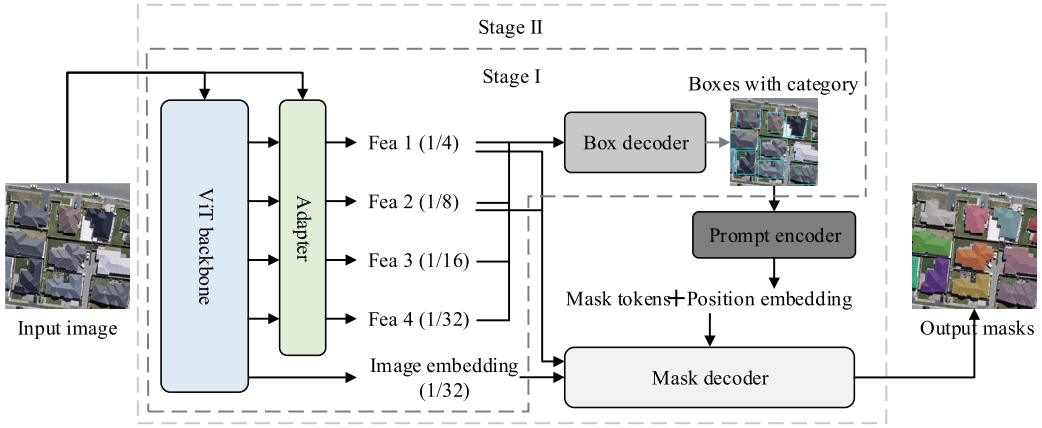


Fig. 2. Overview of the proposed SAM-RSIS. Stage I is enclosed with the dark gray dashed line, and Stage II with the light gray dashed line. Fea1, Fea2, Fea3, and Fea4 are multiscale features output from the adapter.

the semantic segmentation model to better learn the boundary information of objects. GeoSAM [22] fine-tunes SAM's mask decoder with both sparse and dense visual prompting for the mobility infrastructure segmentation task. RSPrompter [23] designs a prompter to generate continuous prompts to replace the original discrete prompts and prompt encoder in SAM, enabling SAM to produce category-specific instance segmentation results for remote sensing images. Osco et al. [57] use the highest scoring object calculated by the grounding-DINO as a reference and use the PerSAM-F to obtain the final segmentation mask. These few studies are mostly designed for the semantic segmentation task [21], [22], only RS-prompter for RSIS. However, as both the ViT backbone and mask decoder of SAM are frozen during training the prompter, RS-prompter faces challenges in handling the significant domain shift between remote sensing images and natural scene images. In this article, we comprehensively investigate each part of SAM and propose a more reasonable and effective fine-tuning method.

III. Methodology

In this section, we introduce how we adapt the original SAM for general segmentation to RSIS. In Section III-A, we give a brief overview of the proposed method SAM-RSIS. We describe the structure of the adapter used to fine-tune the ViT backbone in Section III-B. Subsequently, we provide a detailed description of the two stages of SAM-RSIS in Sections III-C and III-D, respectively.

A. Overview

As shown in Fig. 2, upon the ViT backbone (image encoder), prompt encoder, and mask decoder in the original SAM, we introduce an adapter to alleviate the domain shift between natural scene images and remote sensing images and also tackle the challenge of significant size variations of objects on remote sensing images. We also introduce a box decoder for providing automatic bounding box prompts. During the training phase, the ViT backbone pretrained on the large-scale dataset SA-1B is frozen. The adapter is trained with remote sensing data to output multiscale features more suitable for the remote sensing images. The box decoder then takes the multiscale features as inputs and outputs heatmaps, size maps, and offset maps. The ViT backbone, adapter, and box decoder form an object detector (i.e., the first stage). After training, the object detector provides the position and size information of target objects (i.e., bounding boxes and center points) to the prompt encoder in the next stage.

In the second stage, object masks are generated within the predicted bounding boxes to achieve instance segmentation. In the training phase, we fine-tune the mask decoder pretrained on natural scene images. We freeze the ViT backbone, adapter, and box decoder that have been trained in the first stage and also the weights of the prompt encoder in SAM. The prompt encoder encodes the bounding boxes and center points into positional embeddings. The position embedding, two newly introduced mask output tokens, which are learnable embeddings responsible for predicting masks at 1/4 and 1/8 scales, the low-resolution (1/32) image embedding from the ViT backbone, as well as two features from the adapter (at 1/4 and 1/8 scales), are fed into the mask decoder altogether to predict the masks of objects.

B. Adapter

To adapt the ViT backbone efficiently to the remote sensing domain with affordable trainable parameters and also address

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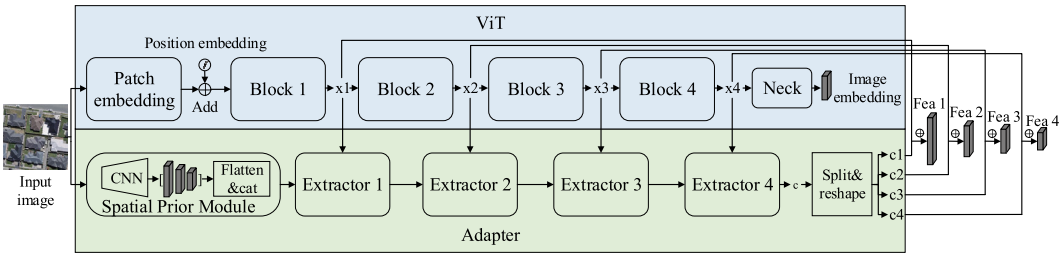


Fig. 3. Adapter structure. x1–x4 are global features output from the four blocks of ViT backbone. c is the output feature from Extractor 4. c1–c4 are multiscale features obtained by splitting the feature c and reshaping. is the position embedding and is the elementwise addition.

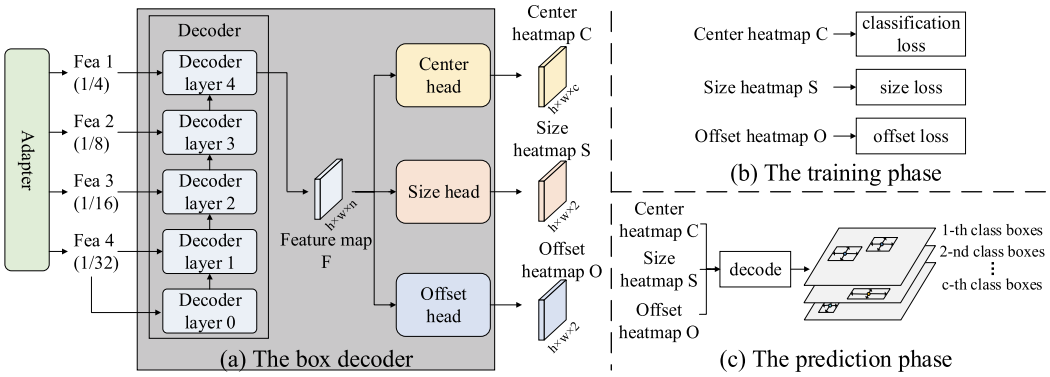


Fig. 4. (a) Structure of box decoder. (b) Losses in the training phase. (c) Operations in the prediction phase.

its shortcomings of only generating 1/16-scale features for small object detection in remote sensing images, we introduce an adapter inspired by ViT-Adapter [58], as shown in Fig. 3. We discard the injector module contained in ViT-Adapter to speed up training with fewer computing resources while preserving the feature distribution and maintaining training stability. Our adapter contains a spatial prior module and four extractors. The spatial prior module is a CNN utilized to model the local spatial information of input images. It ultimately yields a feature vector formed by flattening and concatenating multiscale features. The four extractors extract global features from the four blocks of ViT backbone (ViT-Huge) to interact with the local trainable spatial features in the adapter. Specifically, in each extractor, this interaction is achieved through an attention layer and a feed-forward network (FFN). Cross-attention is a typical way in the modern transformer structure for information interaction. First, the local spatial feature served as the query and the global feature acted as the key and value are input into the attention layer that employs deformable attention [59] for cross-attention.

Then, the output feature is fed into the FFN that contains two fully connected layers, producing the feature with robust representation capability.

The final output feature c from the fourth extractor (Extractor 4 in Fig. 3) in the adapter is split and reshaped into four scale features c1–c4, whose sizes are 1/4, 1/8, 1/16, and 1/32 of the image size, respectively. Finally, the output features of each block in the ViT backbone are resized to the size of the corresponding features ci (i = 1–4) and added to them, respectively, resulting in the final Fea1, Fea2, Fea3, and Fea4 at scales of (1/4, 1/8, 1/16, and 1/32).

C. Stage I: Object Detection

Bounding box is a natural prompt form for instance segmentation. We perform object detection to automatically generate box prompts, thus forcing the mask decoder of SAM to focus on the object regions and generate refined masks. As anchor-based object detection methods predefine a large number of proposal boxes far exceeding the number of targets, we adopt the more efficient object-based strategy to build a box decoder, together with the ViT backbone and adapter to form an object detector.

The box decoder contains a decoder and three prediction heads [Fig. 4(a)]. The decoder first decodes the minimum scale feature Fea4 (1/32) from the adapter and fuses it with the multiscale features Fea3, Fea2, and Fea1 (1/16, 1/8, and 1/4) from the adapter through skip connections progressively. The final generated feature map F has a size of $h \times w \times n$, where h and w are 1/4 size of the height and width of the input image (i.e., 256 pixels), and n represents the number of channels. Each decoder layer consists of the following components: a modulated deformable convolutional layer, followed by a batch normalization layer and a ReLU activation function, then a transpose convolutional layer, followed by another batch normalization layer and a ReLU activation function.

The three prediction heads consist of the center head, size head, and offset head, each consisting of a 3×3 convolutional layer followed by a 1×1 convolutional layer. Specifically, the feature map F outputted by the decoder is inputted into these three prediction heads. The center head outputs a center-point heatmap C of size $h \times w \times c$, where c represents the number of classes. The value at each pixel on each channel represents the probability that the point is the center of an object of each class. The size head outputs a size heatmap S of size

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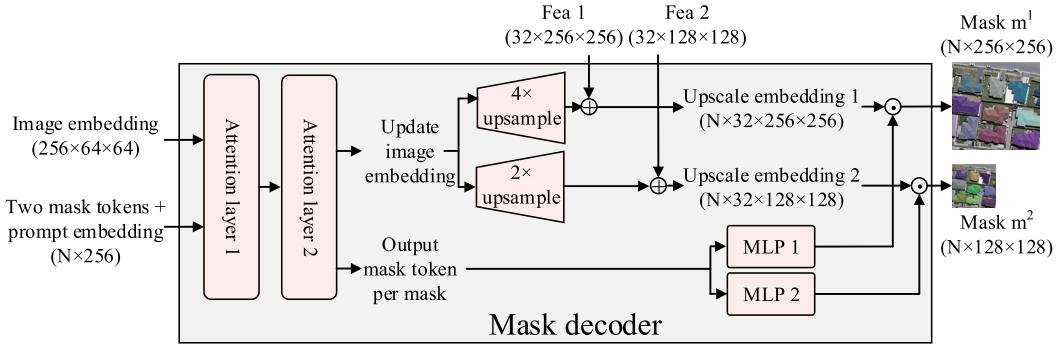


Fig. 5. Structure of the mask decoder.

$h \times w \times 2$, which estimates the size (height and width) of the object centered at each pixel. Similarly, the offset head outputs an offset heatmap O of size $h \times w \times 2$. This offset heatmap O predicts the offset values of the predicted center points relative to the actual object center points in the height and width directions to correct the discretization error caused by image downsampling.

We train the adapter and the box decoder following CenterNet [29] [Fig. 4(b)]. We apply the modified focal loss $\mathcal{L}_{\mathrm{C}}$ to supervise the center-point heatmap C, which is defined as follows:

$$\mathcal{L}_{\mathrm{C}} = -\frac{1}{M}\sum_{i=1}^{h}\sum_{j=1}^{w}\sum_{k=1}^{c}\begin{cases}\left(1-\mathrm{C}_{ijk}\right)^{\alpha}\cdot\log\left(\mathrm{C}_{ijk}\right), & \mathrm{C}_{ijk}^{\mathrm{gt}}=1\\ \left(1-\mathrm{C}_{ijk}^{\mathrm{gt}}\right)^{\beta}\cdot\left(\mathrm{C}_{ijk}\right)^{\alpha}\cdot\log\left(1-\mathrm{C}_{ijk}\right), & \text{otherwise}\end{cases} \tag{1}$$

where C^{gt} is the heatmap of center points of ground-truth objects calculated by Gaussian kernel. M is the number of ground-truth objects. α and β are the weights set to 2 and 4, respectively, followed the setting in the CenterNet [29] and Cornernet [60].

Besides, we use L1 loss to supervise the offset heatmap O and size heatmap S, which are formulated as shown in (2) and (3), respectively. It is worth noting that these two losses are supervised only at the positions of the actual object center points

$$\mathcal{L}_{\mathrm{O}} = \frac{1}{M}\sum_{i=1}^{M}\left|\mathrm{O}_{p_i} - \left(\frac{p_i^{\mathrm{gt}}}{4} - p_i\right)\right| \tag{2}$$

$$\mathcal{L}_{\mathrm{S}} = \frac{1}{M}\sum_{i=1}^{M}\left|\mathrm{S}_{p_i} - \mathrm{S}_{p_i}^{\mathrm{gt}}\right| \tag{3}$$

where $p^{\mathrm{gt}} \in \mathbb{R}^2$ is the coordinate of actual object center points in the original image size, p is the coordinate of actual object center points in O that is 1/4 the size of the original image, and $\mathrm{S}_p^{\mathrm{gt}}$ is the actual size of objects at coordinates p.

The overall loss function is as follows:

$$\mathcal{L}_{\mathrm{det}} = \mathcal{L}_{\mathrm{C}} + \mathcal{L}_{\mathrm{O}} + 0.1\mathcal{L}_{\mathrm{S}}. \tag{4}$$

In the prediction phase [Fig. 4(c)], for generating bounding boxes of target objects, the information in the center map C, size map S, and offset map O are gathered and decoded. First, we obtain the top K points with the highest scores from the center heatmap C. These points are regarded as the centers of objects. Then, the position of the center point is corrected through offset map O. Finally, the top-left and bottom-right corner coordinates of the bounding box of detected objects are calculated through the center point position and the corresponding width and height values on size map S.

D. Stage II: Mask Generation

In the mask generation stage, the key structure is the mask decoder. Unlike the ViT backbone with huge parameters, the mask decoder in SAM is a lightweight transformer and the entire structure can be fully retrained to adapt to the RSIS task.

We make two modifications to the original mask decoder of SAM, as shown in Fig. 5. First, we design two high-quality mask output tokens to replace three mask output tokens and one IoU token in the original SAM. Second, we introduce Fea1 and Fea2 outputted from the adapter, which are of sizes 256 × 256 and 128 × 128, respectively, into the mask decoder. There are several reasons for the above modifications. First, we no longer need multiple mask tokens and IoU tokens to address segmentation ambiguities caused by unclear prompts. We have explicitly used the predicted bounding boxes as prompts. Second, we consider the use of multiscale high-resolution masks can handle the significant size variations of objects in remote sensing images and obtain finer mask boundaries. Additionally, Fea1 and Fea2 themselves contain rich semantic contextual information.

In the training phase, we use Hungarian algorithm [61] to match the prediction masks at two scales, denoted as $m^1 = \{m_i^1 | m_i^1 \in [0,1]^{256\times256}\}_{i=1}^{N}$ and $m^2 = \{m_i^2 | m_i^2 \in [0,1]^{128\times128}\}_{i=1}^{N}$), with the ground truth masks $m^{\mathrm{gt}} = \{m_i^{\mathrm{gt}} | m_i^{\mathrm{gt}} \in [0,1]^{H\times W}\}_{i=1}^{M}$, respectively. N and M are the number of prediction masks at each scale and ground truth instance masks (N ≥ M). We define the matching cost $\hat{\sigma}$ at two scales as in the following equation:

$$\hat{\sigma}^j = \underset{\sigma^j\in\mathfrak{S}_N}{\arg\min}\sum_{i}^{N}\left[\mathcal{L}_{\mathrm{ce}}\left(m_i^j, m_{\sigma^j(i)}^{\mathrm{gt}}\right) + \mathcal{L}_{\mathrm{dice}}\left(m_i^j, m_{\sigma^j(i)}^{\mathrm{gt}}\right)\right] \tag{5}$$

where $j = \{1, 2\}$ represents the scales 256 × 256 and 128 × 128, respectively. σ represents the mapping relationship. $\mathcal{L}_{\mathrm{ce}}(\cdot)$ and $\mathcal{L}_{\mathrm{dice}}(\cdot)$ are the binary cross-entropy loss and

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dice loss, respectively. After obtaining a matching, we compute the mask loss, which consists of the binary cross-entropy loss and dice loss for each predict instance mask per scale, which can be denoted as follows:

$$\mathcal{L}^{j}_{\text{mask}} = \sum_{i}^{N} \left[\mathbb{1}\mathcal{L}_{\text{ce}}\left(m_i^j, m^{\text{gt}}_{\sigma^j(i)}\right) + \mathbb{1}\mathcal{L}_{\text{dice}}\left(m_i^j, m^{\text{gt}}_{\sigma^j(i)}\right) \right] \quad (6)$$

where $\mathbb{1}$ is the indicator function confirming a positive match. Therefore, the loss function in the mask generation stage is shown in following equation:

$$\mathcal{L}_{\text{seg}} = \mathcal{L}^{1}_{\text{mask}} + \mathcal{L}^{2}_{\text{mask}}. \quad (7)$$

For efficiently training the mask decoder, in matching, and the final loss calculation, we compute the loss on R random sample points instead of the entire mask, using the strategy in Mask2Former [35]. Following Mask2Former, we set R to 12 544.

During the prediction phase, we up sample the two-scale masks to the size of original image and numerically average them to obtain the final prediction instance masks.

IV. Experiments

A. Datasets

To assess both the performance and generalization ability of the proposed method, we selected three datasets for experiments, i.e., the WHU aerial building dataset [62], the WHU-Mix (vector) building extraction dataset [37], and the NWPU VHR-10 dataset [63]. These three datasets can reflect different real-world scenarios of RSIS. The WHU dataset and WHU-Mix dataset are both single-class instance segmentation datasets, focusing on the extraction of buildings. The WHU dataset contains images captured in the same area. The WHU-Mix dataset is much larger than the WHU dataset, with training and testing samples from different and various places worldwide, making it very challenging. NWPU is designed for the multiclass segmentation task.

1) WHU Aerial Building Dataset: The WHU aerial building dataset [62] is a widely used high-quality remote sensing building extraction dataset. The aerial images were captured over the city of Christchurch, New Zealand, covering an area of about 200 km^2 with a resolution of 0.2 m. The dataset offers high-precision building labels in both raster and vector formats, with vector-based labels stored in SHP (Shapefile) format. We followed the processing flow of Wei et al. [37], transforming the vector label format from SHP to MS-COCO-style JSON. We cropped the images into patches with the size of 1024 × 1024 pixels, in which seamless partitioning is used for the training set and validation set, and overlapping partitioning is used for the test set. As a result, 2793, 627, and 2220 tiles were obtained in the training set, validation set, and test set.

2) WHU-Mix (Vector) Building Extraction Dataset: The WHU-Mix (vector) building extraction dataset [37] is a large-scale and diverse building extraction dataset, which consists of about 64 000 image tiles captured from different sources and places all over the world and covers an area of about 1100 km^2. The images in the WHU-Mix dataset have three sizes, namely, 300 × 300 pixels, 512 × 512 pixels, and 650 × 650 pixels. The building instance labels are stored in MS-COCO-style JSON format. The dataset is divided into four sets, i.e., the training set (43 778 tiles), the validation set (2922 tiles), the test set I (11 675 tiles), and the test set II (6011 tiles). Specially, there is no geographical overlap between the images in test set II and the training set, so we can evaluate the generalization ability and robustness of a model toward the realistic real-world case with test set II.

3) NWPU VHR-10 Dataset: The NWPU VHR-10 dataset is a multiclass remote sensing object detection dataset [63]. Ten classes are annotated: airplane (ai), ship (sh), storage tank (st), baseball diamond (bd), tennis court (tc), basketball court (bc), ground track field (gt), harbor (ha), bridge (br), and vehicle (ve). 715 images were collected from Google Earth with spatial resolution ranging from 0.5 to 2 m, and 85 images were from ISPRS Vaihingen dataset [64] with a resolution of 0.08 m. The image sizes range from 500 to 1100 pixels in length and width. The dataset divided the images into a positive image set and a negative image set. The positive set comprises images with one or more ground objects, while the negative image set contains images without ground objects, totaling 650 and 150 images, respectively. Su et al. [65] provided the instance annotations for the positive image set in the NWPU dataset. We follow the setting in RSPrompter [23] and use 80% of the data in the positive image set for training and the remaining 20% for testing.

B. Experimental Settings and Evaluation Metrics

1) Experimental Settings: All experiments conducted in this study were implemented in PyTorch on a computer with two NVIDIA RTX A6000 48-GB GPUs and an Intel Xeon Silver 4109T CPU. For the proposed SAM-RSIS, we employed the ViT-Huge as the image encoder. The weights of ViT-Huge, prompt encoder, and mask decoder in the SAM-RSIS were initialized using the weights of the original SAM [16], and then, the weights of ViT-Huge and prompt encoder were frozen during the entire training process. In the box generation stage, the adapter and box decoder were trained by the AdamW optimizer [66], with an initial learning rate of 1×10^{-5} for the WHU-Mix dataset and of 1×10^{-4} for the WHU and NWPU datasets and a weight decay of 1×10^{-5}. Using the multistep learning rate scheduler, the WHU and WHU-Mix datasets were trained for a maximum of 15 000 iterations, while the NWPU dataset was trained for a maximum of 9000 iterations. Steps were set at 70% and 90% of the maximum iterations, with a gamma value of 0.1. A warm-up of 100 iterations was performed. In the mask generation stage, the mask decoder was trained by the AdamW optimizer, setting the initial learning rate to 1×10^{-4} and the weight decay to 1×10^{-5} for all datasets. The learning rate scheduler was the same to that of the box generation stage. During training, we set the input image size to 1024 × 1024, consistent with SAM, and images with inconsistent sizes were resized to match this size. Besides, the random horizontal flipping and random color jittering were used as data augmentation. To fit the GPU capacity, the batch size was set to 10 and 4 in the box generation stage and mask generation stage, respectively.

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TABLE I
QUANTITATIVE COMPARISON ON THE WHU AERIAL BUILDING DATASET

Method		Backbone	mAP	AP_{50}	AP_{75}
The advanced instance segmentation methods	Mask R-CNN	Resnet 50	69.1	90.7	81.6
	MS R-CNN	Resnet 50	69.3	90.5	81.7
	YOLACT	Resnet 50	43.2	69.3	50.7
	SOLO v2	Resnet 50	56.0	82.8	64.0
	CondInst	Resnet 50	69.0	**91.3**	80.5
	Mask2Former	Resnet 50	67.6	83.8	76.7
	BuildMapper	Resnet 50	**74.6**	89.0	83.5
The derivatives of SAM	SAM-Seg	ViT-Huge	68.6	89.1	79.7
	SAM-Point	ViT-Huge	33.5	50.6	40.2
	SAM-Box	ViT-Huge	60.0	86.2	69.1
	RSPrompter	ViT-Huge	60.5	83.1	70.1
	Grounded-SAM	ViT-Huge	34.5	50.8	39.2
	ours	ViT-Huge	72.6	91.0	**84.2**

2) Evaluation Metrics: We employed the standard COCO mean average precision (mAP) to quantitatively assess instance segmentation results from different methods across the test sets of three datasets. The mAP is the average precision (AP) over ten mask IoU thresholds (i.e., from 0.50 to 0.95 with steps of 0.05) and all classes. Here, IoU is the ratio between the intersection area of predicted mask and ground truth mask and their union area.

C. Comparison to the State of the Art

1) Comparison Methods: We compare the performance of the proposed method with both the advanced instance segmentation methods and the derivatives of SAM.

The general instance segmentation methods are mask R-CNN [5], MS R-CNN [25], YOLACT [6], SOLO V2 [31], CondInst [33], and Mask2Former [35]. Among them, mask R-CNN and MS R-CNN are classical two-stage anchor-based methods, YOLACT is a one-stage anchor-based method, while SOLO V2 and CondInst are one-stage anchor-free methods, and Mask2Former is a SOTA transformer-based method. For WHU and WHU-Mix datasets, we also compare with BuildMapper [37], which is a SOTA building extraction method that achieves polygonal building extraction through learnable contour initialization and evolution.

Furthermore, we compare five derivative products of SAM.

1) *SAM-Seg* replaces the backbone of Mask2Former with pretrained ViT backbone of SAM and trains the remaining parts of Mask2Former.
2) SAM-Point uses the 32 × 32 grid points to prompt SAM, where the points falling within instance masks are considered positive mask points, while the remaining points are background points. Additionally, a class token is introduced to learn the class of points.
3) SAM-Box utilizes boxes generated by a pretrained object detector Centernet [29] (Resnet 50 as the backbone) on the experimental dataset to prompt the original SAM to generate corresponding masks.
4) RSPrompter [23] automatically prompts SAM using learned prompt embeddings. It is worth noting that the training of this method requires a large GPU memory and a long convergence time. Under our experimental condition, the batch size could only be set to 8. For the WHU, WHU-Mix, and NWPU datasets, we conducted training for 330, 40, and 600 epochs, respectively. This resulted in training times of about 5, 7, and 1.5 days, respectively (the training times of our method are about 2.6, 2.8, and 1.3 days).
5) Grounded-SAM [54] is a zero-shot method that uses the open-world object detector Grounding-DINO [55] to generate boxes to prompt SAM. We applied the parameter settings of grounded-SAM [54] and set the box and text thresholds for grounding-DINO to 0.3 and 0.25, respectively. Additionally, for the WHU and WHU-Mix datasets, we used "house" as the text prompt for grounding-DINO, as we found that this produced better results compared to using "building." For the NWPU dataset, we used the class names defined in the dataset to prompt grounding-DINO, respectively. Then, the segmentation results of all classes were aggregated to obtain the final instance segmentation results.

2) Results for the WHU Dataset: Table I presents the quantitative results on the WHU dataset. According to Table I, our method exhibits competitive performance across all metrics. Specifically, our method achieves a mAP score of 72.6, ranking second, slightly lower than BuildMapper; the latter is specifically designed for building instance segmentation tasks. Notably, our method outperforms all comparative methods in terms of AP_{75}, suggesting its proficiency in capturing fine object contour details and achieving high-quality instance segmentation. Among the advanced instance segmentation methods, most achieve satisfactory performance. YOLACT and SOLO V2, as single-stage methods, compromise some accuracy for efficiency. For the derivatives of SAM, the performance varies depending on the specific modifications and adjustments made to the original SAM. The lower accuracy of SAM-Point may be attributed to the use of points as prompts, which may not provide sufficient detailed information about instances. On the other hand, grounded-SAM leverages frozen grounding-DINO and SAM models. While both grounding-DINO and SAM possess some degree of generalization ability, they are limited when faced with remote sensing images that exhibit obvious domain shifts from natural scene images.

Fig. 6 presents an example of the instance segmentation results from the WHU dataset. We can observe that both our

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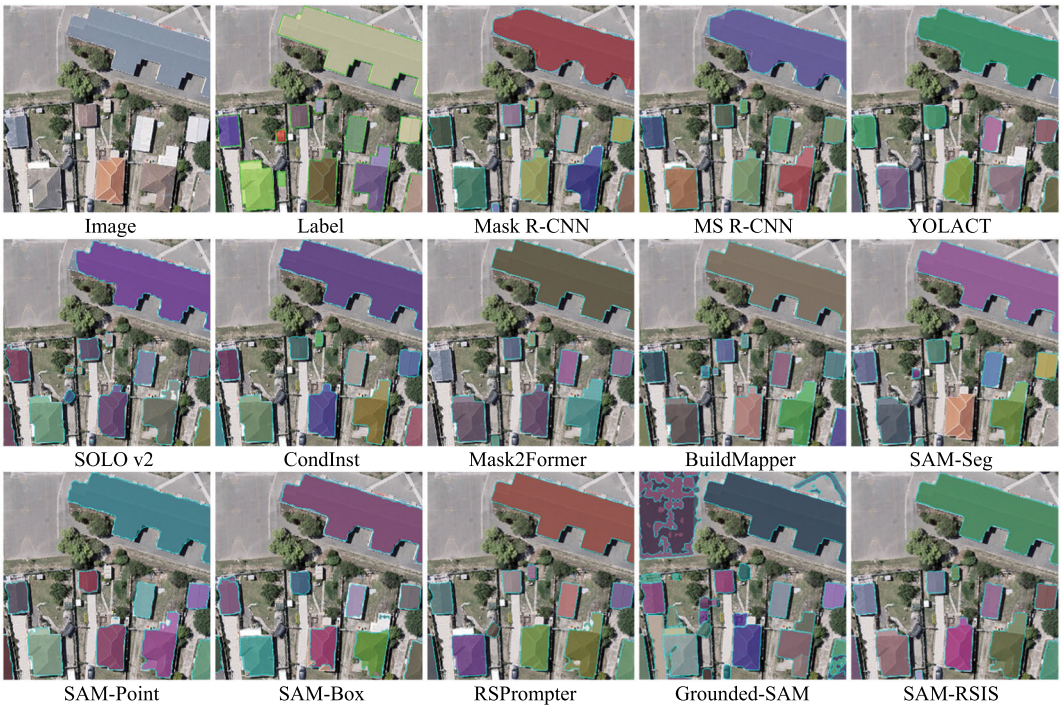


Fig. 6. Prediction results of the different methods for a 300 × 300 image patch in the WHU dataset. In the prediction results and label, contours of buildings are depicted in cyan and green, respectively, and different building instances are represented by different color masks.

proposed method and BuildMapper are capable of producing clear and sharp edges of building instances. However, mask R-CNN, MS R-CNN, YOLACT, SOLO v2, and SAM-Point struggle to handle the fine details of large building contours. Numerous noisy masks are observed in the results of grounded-SAM.

3) Results for the WHU-Mix Dataset: The experimental results for the WHU-Mix dataset are listed in Table II. For the WHU-Mix test set I, the proposed SAM-RSIS achieves the mAP and AP scores on par with the advanced instance segmentation methods; however, there is still a performance gap of 3% mAP compared to the best-performing method, MS R-CNN. It is not surprising that a well-designed instance segmentation method with thorough training on the large WHU-Mix dataset performs well on test set I, which shares the same distribution with the training set. In contrast, the frozen ViT-Huge backbone in SAM cannot fully benefit from a large target training data. Nevertheless, our method ranks the first place among those derivatives of SAM, indicating the effectiveness of our fine-tuning strategy.

For the more challenging WHU-Mix test set II, which has no geographically adjacent regions to the training set, our method achieves the best performance across all metrics compared to all the other methods. Those advanced instance segmentation methods, such as Mask2Former, lose their advantages of powerful interpolation ability, because in this time, extrapolation ability is examined. Through an exquisite design, our method can make full use of the generalization ability that is embedded in the SAM backbone pretrained on extremely large data. Furthermore, our method only experiences a slight decrease of 3.5% in mAP: from 55.9% on test set I to 52.4% on test set II. In contrast, the performance of advanced instance segmentation methods exhibits a significant drop on this test set. For instance, the mAP of the top-performing MS R-CNN on test set I declines from 58.9% to 48.9%. These above results demonstrate the robustness and great generalization ability of our method, while the compared instance segmentation methods struggle to maintain stability when dealing with cross-domain scenarios. The performance of compared derivatives of SAM is not very satisfactory, indicating their designs are not optimal for applying SAM to remote sensing images. Unlike other methods, grounded-SAM does not experience a decrease in accuracy between test sets I and II. This is because grounded-SAM does not require training, both the WHU-Mix test sets I and II represent new domains for this method.

A visual comparison of different methods on the WHU-Mix tests I and II is shown in Figs. 7 and 8, respectively. It can be seen that our method, SAM-Seg, and most of the compared instance segmentation methods can generate relatively accurate masks with little noise for the building instances of the WHU-Mix test I data. However, for the WHU-Mix test II data, the instance segmentation quality of most methods is not ideal. Our method can still identify almost all the buildings accurately.

4) Results for the NWPU Dataset: We report the results obtained on the NWPU dataset in Table III. The proposed SAM-RSIS achieves the highest mAP score of 64.1%, demonstrating success in handling multiclass instance segmentation tasks in remote sensing. SAM-RSIS also outperforms all comparison methods in mIoU and AP75. Additionally, it achieves the highest AP scores for six categories, indicating its capability to address issues related to class similarity and class imbalance effectively. Interestingly, compared to the single-class tasks mentioned earlier, some instance

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TABLE II

QUANTITATIVE COMPARISON ON TEST SETS I AND II OF THE WHU-MIX (VECTOR). TEST SET II REFLECTS CROSS-DOMAIN AND REAL-WORLD CHALLENGES THUS RESULT ON IT ARE MORE CONVINCING

Test set			Test set I			Test set II		
Method		Backbone	mAP	AP_{50}	AP_{75}	mAP	AP_{50}	AP_{75}
The advanced instance segmentation methods	Mask R-CNN	Resnet 50	58.2	**85.1**	66.6	50.2	80.1	53.8
	MS R-CNN	Resnet 50	**58.9**	84.9	**67.1**	48.9	77.8	52.8
	YOLACT	Resnet 50	51.0	77.9	58.0	43.0	72.3	44.9
	SOLO v2	Resnet 50	56.4	83.5	63.3	48.8	78.5	51.7
	CondInst	Resnet 50	56.6	83.7	63.6	47.4	78.4	49.8
	Mask2Former	Resnet 50	52.0	77.9	58.2	44.7	72.7	46.7
	BuildMapper	Resnet 50	58.0	82.6	65.9	48.1	73.2	52.0
The derivatives of SAM	SAM-Seg	ViT-Huge	54.4	80.5	60.5	49.5	77.8	52.6
	SAM-Point	ViT-Huge	21.2	37.8	22.3	22.6	40.7	23.5
	SAM-Box	ViT-Huge	47.7	80.5	50.7	44.3	74.9	46.5
	RSPrompter	ViT-Huge	46.3	77.9	49.2	43.8	74.2	45.7
	Grounded-SAM	ViT-Huge	15.7	27.9	15.7	20.6	37.9	19.9
	ours	ViT-Huge	55.9	81.9	62.9	**52.4**	**80.2**	**56.9**

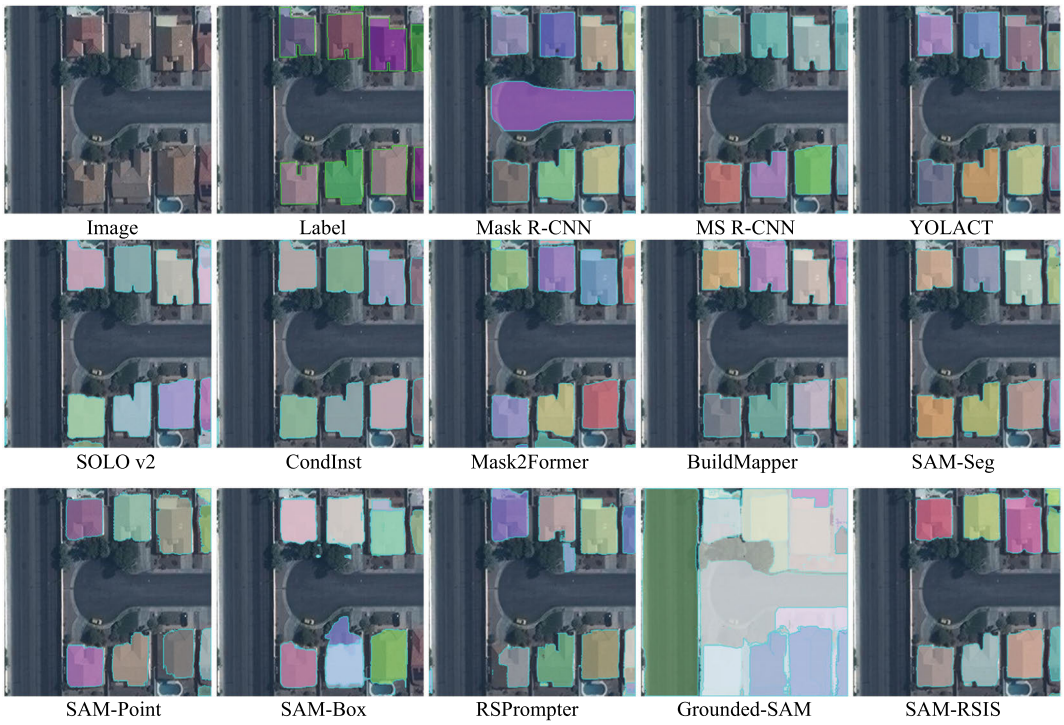


Fig. 7. Prediction results of different methods in test set I of the WHU-Mix dataset.

segmentation methods exhibit noticeable performance degradation when faced with the more complex multiclass task on the NWPU dataset. For example, while MS R-CNN ranks third and first on the WHU and WHU-Mix test set I, respectively, its mAP ranking on the NWPU dataset is only 8 out of 12. In contrast, our method demonstrates strong stability, yielding satisfactory performance across all three datasets. Among the derivatives of SAM, SAM-Seg remains the best apart from our method. Grounded-SAM fails to produce meaningful results. This could be attributed to significant appearance differences in these categories between natural and remote sensing images.

Fig. 9 gives an image patch example from the NWPU validation set. It can be observed that our method produces satisfactory segmentation masks for each instance. However, many methods struggle to accurately distinguish each instance, especially for instances of the "tennis court" class, resulting in a large number of erroneous masks.

D. Ablation Studies

To further evaluate the effectiveness of different modules and strategies used in the SAM-RSIS, we perform ablation studies in the three experimental datasets. The results are listed in Table IV. "All modules are unused" refers to employing the frozen ViT-Huge backbone of SAM for object detection and only training the box decoder to prompt SAM for generating masks. "Adapter" involves introducing the adapter to form our object detector to prompt SAM. "Fine-tuning the mask

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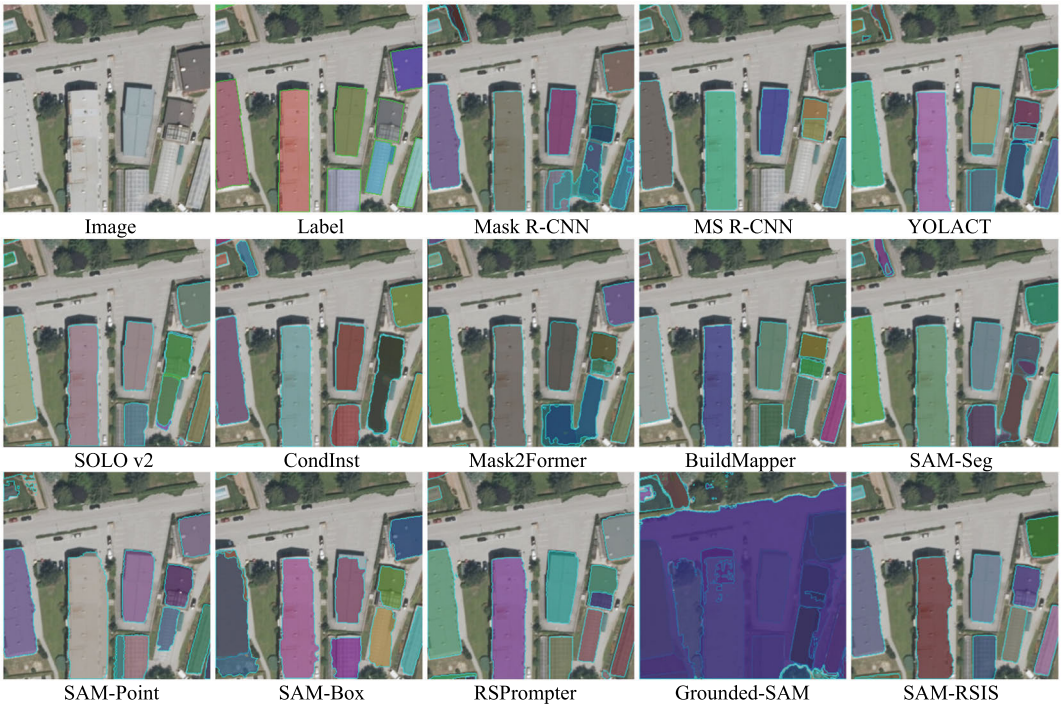


Fig. 8. Prediction results of different methods in test set II of the WHU-Mix dataset.

TABLE III

QUANTITATIVE COMPARISON ON THE NWPU DATASET. THE ABBREVIATIONS ARE: AIRPLANE (AI), SHIP (SH), STORAGE TANK (ST), BASEBALL DIAMOND (BD), TENNIS COURT (TC), BASKETBALL COURT (BC), GROUND TRACK FIELD (GT), HARBOR (HA), BRIDGE (BR), VEHICLE (VE), RESNET (R50), AND VIT-HUGE (VIT-H)

Method		Backbone	mAP	AP^{50}	AP^{75}	AP^{ai}	AP^{sh}	AP^{st}	AP^{bd}	AP^{tc}	AP^{bc}	AP^{gt}	AP^{ha}	AP^{br}	AP^{ve}
The advanced instance segmentation methods	Mask R-CNN	R50	45.8	80.6	45.3	21.8	48.1	82.9	75.1	43.4	42.9	53.8	33.2	17.9	38.6
	MS R-CNN	R50	41.9	72.3	43.6	20.0	45.2	80.0	73.5	46.9	30.0	59.8	20.7	8.2	34.1
	YOLACT	R50	37.6	66.6	38.8	3.2	33.6	73.6	75.1	19.8	27.5	81.9	27.9	11.7	21.2
	SOLO v2	R50	41.0	69.3	42.1	26.1	32.4	49.1	66.5	38.6	51.7	8.9	18.1	17.6	21.3
	CondInst	R50	62.8	**89.8**	66.2	41.5	55.2	**84.1**	81.7	67.8	84.2	94.9	43.0	26.0	52.6
	Mask2Former	R50	57.8	80.8	62.2	42.7	54.3	83.1	76.1	61.6	69.7	87.9	38.8	29.7	34.2
The derivatives of SAM	SAM-Seg	ViT-H	58.0	87.7	61.9	24.4	53.9	74.3	83.6	63.1	68.3	94.1	**44.2**	**36.5**	37.6
	SAM-Point	ViT-H	23.8	35.9	25.0	9.5	23.8	18.9	52.0	18.9	14.4	76.5	12.4	2.7	8.9
	SAM-Box	ViT-H	46.0	73.8	49.5	14.4	55.9	78.7	61.3	68.6	67.5	42.4	26.3	1.6	43.1
	RSPrompter	ViT-H	51.7	76.8	53.5	16.1	53.6	78.8	79.3	61.1	52.9	**96.9**	30.8	17.4	29.5
	Grounded-SAM	ViT-H	1.7	2.7	1.7	1.1	4.8	0.1	3.3	0.1	0.2	4.9	0.1	0.1	2.2
	ours	ViT-H	**64.1**	88.5	**68.8**	**50.3**	**61.6**	84.0	**85.6**	**77.6**	**71.4**	95.2	35.7	23.0	**56.4**

decoder" means training and fine-tuning SAM's mask decoder using the box prompts generated by the trained object detector, but with only one mask output token used to output a scale of masks. "Multilevel feature" denotes using a multiscale strategy during the fine-tuning of SAM's mask decoder to output masks at two scales.

Similar conclusions can be drawn from the results of the three datasets. First, the result of "all modules are unused" is worst, indicating that SAM trained on natural scene images cannot be effectively applied to remote sensing images instance segmentation task even though a box decoder is trained on new data. Second, the separate use of three modules all leads to performance improvements, and the combination of these modules achieves the best results. Specifically, the gain from "adapter" is relatively small, which is understandable because the parameters of ViT-Huge can only be fine-tuned indirectly through the adapter. The largest performance gain comes from "fine-tuning the mask decoder," highlighting the importance of fine-tuning the model to adapt to remote sensing data and handle instance segmentation task. "Multilevel feature," building upon "fine-tuning the mask decoder," has a moderate effect by addressing the issue of significant variations of object sizes in remote sensing images with a high-resolution multiscale strategy.

V. DISCUSSION

The above results and observations bring us some insights that not clearly reported in previous studies attempting to fine-tune SAM to remote sensing image segmentation. First, adjusting both the front-end ViT backbone and back-end mask decoder is important to handle the domain shift

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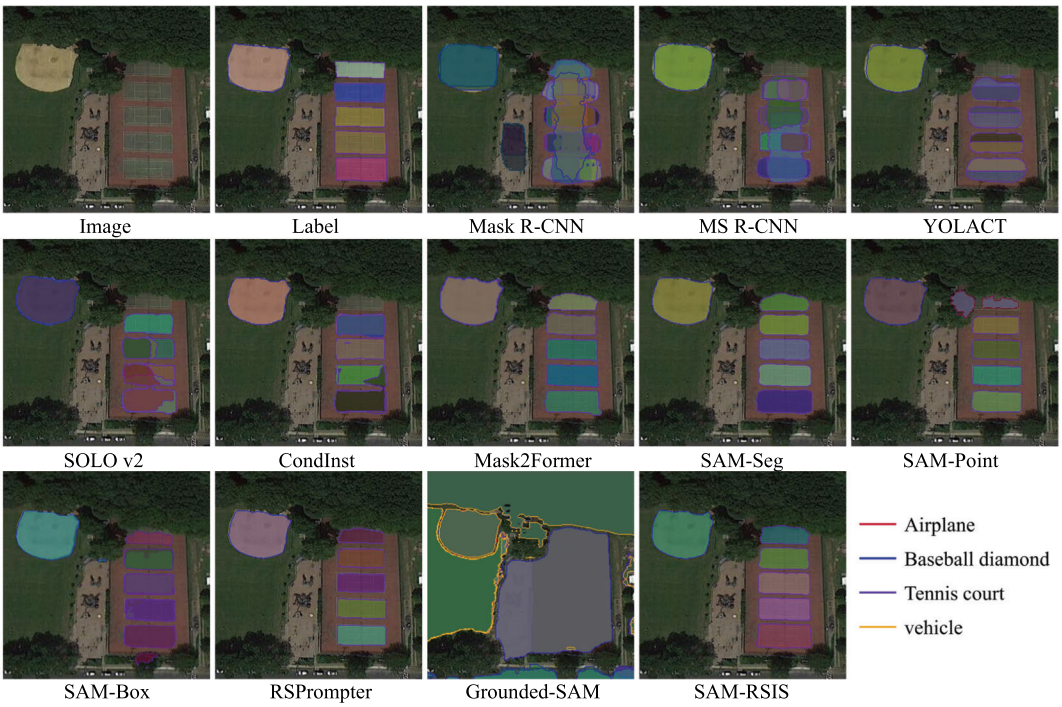


Fig. 9. Prediction results of different methods in the NWPU dataset. In both the prediction results and label, contours of different classes are depicted with distinct colors in the legend, and different instances are represented by different color masks.

TABLE IV
ABLATION STUDIES ON THE WHU, WHU-MIX, AND NWPU DATASETS

Dataset			WHU			WHU-Mix test set I			WHU-Mix test set II			NWPU		
Module			Metrics											
Adapter	Fine-tune the mask decoder	Multi-level feature	mAP	AP_{50}	AP_{75}	mAP	AP_{50}	AP_{75}	mAP	AP_{50}	AP_{75}	mAP	AP_{50}	AP_{75}
			60.7	87.5	69.7	45.9	77.3	48.4	45.8	76.9	48.3	47.0	75.8	51.0
✔			62.0	89.4	71.3	46.8	78.6	49.5	46.3	77.3	49.2	48.0	76.3	51.7
✔	✔		71.8	90.4	83.3	53.6	81.3	59.7	51.1	79.6	55.4	61.7	88.0	64.4
✔	✔	✔	72.6	91.0	84.2	55.9	81.9	62.9	52.4	80.2	56.9	64.1	88.5	68.8

problem between remote sensing images and close-range images. Ignoring both of them (e.g., grounded-SAM [54]) will result in bad performance. Only tuning the segmentation decoder (e.g., SAM-Seg) or modifying the prompt encoder (e.g., RSPromper [23]) have limited effects. According to the ablation studies of our method (Table IV), the combination of adjusting the ViT backbone with the adapter and fine-tuning the mask decoder leads to a significant improvement in accuracy and surpassing other derivation methods of SAM.

Second, bounding boxes are an appropriate form for prompting an RSIS task. Through comparing SAM-Point, SAM-Box, and RSPrompter (implicit prompts) across three datasets, we can see that SAM-Point shows the lowest accuracy, while RSPrompter and SAM-Box perform similarly. However, training the prompter with the frozen ViT backbone and mask decoder to obtain implicit prompts involves significantly higher complexity and time costs compared to training a simple object detector to generate box prompts. Therefore, using box prompts is more cost-effective than implicit prompts.

Third, we see that the proposed SAM-RSIS achieves accuracy on par with the state-of-the-art instance segmentation methods, such as Mask2Former [35] or BuildMapper [37] when training and testing sets are similar, and achieves the best performance in cross-domain and more real-world scenarios. This is an encouraging discovery, because in remote sensing applications, the generalization ability and extrapolation ability are much more important for downstream applications, and coincidentally, a large foundation model like SAM is proved meeting such requirements. We regard pretrained foundation models will play their role in more various remote sensing tasks as general backbones.

SAM-RSIS is tuned through two stages, that is to say, when we train parameters in the second stage, we freeze the parameters having been trained in the first stage. Although this pipeline reduces the memory requirements and enhances training stability, the effective parameter updating in the mask decoder cannot be fed back to the adapter. We have experimented with joint two-stage training, and the mAP reached 73.7% on the WHU dataset, which is 1.1% higher than the

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current SAM-RSIS with two-stage separate training. However, gradient explosion occurred during training on the WHU-Mix dataset and NWPU dataset. We think that this problem is mainly due to the poor quality of the boxes generated by the box decoder in the early stage of training. In the future, we will try stabilizing gradient backpropagation with ground truth box prompting in the early epochs. Additionally, we will explore the use of lightweight SAM frameworks, such as MobileSAM [67] and fastSAM [68] as alternatives to the original SAM to alleviate the memory constraint of join training.

VI. Conclusion

This article attempts to apply pretrained SAM to the instance segmentation task in the field of remote sensing. We have proposed a framework named SAM-RSIS, consisting of the box generation stage and the mask generation stage. In the first stage, we construct an object detector upon the ViT backbone of SAM and an introduced adapter. In the second stage, we leverage the bounding box information, combined with the multiscale features of the adapter, to fine-tune the mask decoder. Through the sequential training in the two stages, the SAM model is gradually adapted to remote sensing image data. Extensive experiments on three datasets have demonstrated our method not only outperforms other derivatives of SAM but also shows similar performance to the state-of-the-art instance segmentation methods in "interpolation" scenes and better performance and generalization capability in "extrapolation" scenes. In summary, this article makes a comprehensive exploration to transfer the whole SAM pipeline to the remote sensing domain and will inspire future-related research.

References

[1] S. Qurratulain, Z. Zheng, J. Xia, Y. Ma, and F. Zhou, "Deep learning instance segmentation framework for burnt area instances characterization," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 116, Feb. 2023, Art. no. 103146.

[2] R. Alshehhi, P. R. Marpu, W. L. Woon, and M. D. Mura, "Simultaneous extraction of roads and buildings in remote sensing imagery with convolutional neural networks," *ISPRS J. Photogramm. Remote Sens.*, vol. 130, pp. 139–149, Aug. 2017.

[3] J. Dai, K. He, and J. Sun, "Convolutional feature masking for joint object and stuff segmentation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2015, pp. 3992–4000.

[4] P. O. Pinheiro, R. Collobert, and P. Dollár, "Learning to segment object candidates," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2015, pp. 1990–1998.

[5] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[6] D. Bolya, C. Zhou, F. Xiao, and Y. J. Lee, "YOLACT: Real-time instance segmentation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9156–9165.

[7] H. Su et al., "HQ-ISNet: High-quality instance segmentation for remote sensing imagery," *Remote Sens.*, vol. 12, no. 6, p. 989, Mar. 2020.

[8] Z. Chen, Y. Shang, A. Python, Y. Cai, and J. Yin, "DB-BlendMask: Decomposed attention and balanced BlendMask for instance segmentation of high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5615915.

[9] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[10] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," in *Proc. Int. Conf. Learn. Represent.*, 2021.

[11] J. Deng, W. Dong, R. Socher, L.-J. Li, K. Li, and L. Fei-Fei, "ImageNet: A large-scale hierarchical image database," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, Jun. 2009, pp. 248–255.

[12] O. Russakovsky et al., "ImageNet large scale visual recognition challenge," *Int. J. Comput. Vis.*, vol. 115, no. 3, pp. 211–252, Dec. 2015.

[13] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, "A simple framework for contrastive learning of visual representations," in *Proc. Int. Conf. Mach. Learn.*, 2020, pp. 1597–1607.

[14] K. He, H. Fan, Y. Wu, S. Xie, and R. Girshick, "Momentum contrast for unsupervised visual representation learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 9726–9735.

[15] K. He, X. Chen, S. Xie, Y. Li, P. Dollár, and R. Girshick, "Masked autoencoders are scalable vision learners," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 15979–15988.

[16] A. Kirillov et al., "Segment anything," in *Proc. IEEE Int. Conf. Comput. Vis.*, Oct. 2023, pp. 3992–4003.

[17] M. Oquab et al., "DINOv2: Learning robust visual features without supervision," 2023, *arXiv:2304.07193*.

[18] X. Wang, X. Zhang, Y. Cao, W. Wang, C. Shen, and T. Huang, "SegGPT: Towards segmenting everything in context," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2023, pp. 1130–1140.

[19] S. Gong et al., "3DSAM-adapter: Holistic adaptation of SAM from 2D to 3D for promptable medical image segmentation," 2023, *arXiv:2306.13465*.

[20] W. Ji, J. Li, Q. Bi, T. Liu, W. Li, and L. Cheng, "Segment anything is not always perfect: An investigation of SAM on different real-world applications," 2023, *arXiv:2304.05750*.

[21] X. Ma, Q. Wu, X. Zhao, X. Zhang, M.-O. Pun, and B. Huang, "SAM-assisted remote sensing imagery semantic segmentation with object and boundary constraints," 2023, *arXiv:2312.02464*.

[22] R. Ibn Sultan, C. Li, H. Zhu, P. Khanduri, M. Brocanelli, and D. Zhu, "GeoSAM: Fine-tuning SAM with sparse and dense visual prompting for automated segmentation of mobility infrastructure," 2023, *arXiv:2311.11319*.

[23] K. Chen et al., "RSPrompter: Learning to prompt for remote sensing instance segmentation based on visual foundation model," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701117.

[24] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[25] Z. Huang, L. Huang, Y. Gong, C. Huang, and X. Wang, "Mask scoring R-CNN," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6402–6411.

[26] T. Cheng, X. Wang, L. Huang, and W. Liu, "Boundary-preserving mask R-CNN," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 660–676.

[27] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.

[28] Z. Tian, C. Shen, H. Chen, and T. He, "FCOS: Fully convolutional one-stage object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9626–9635.

[29] X. Zhou, D. Wang, and P. Krähenbühl, "Objects as points," 2019, *arXiv:1904.07850*.

[30] X. Wang, T. Kong, C. Shen, Y. Jiang, and L. Li, "SOLO: Segmenting objects by locations," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 649–665.

[31] X. Wang, R. Zhang, T. Kong, L. Li, and C. Shen, "SOLOv2: Dynamic and fast instance segmentation," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 17721–17732.

[32] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 779–788.

[33] Z. Tian, C. Shen, and H. Chen, "Conditional convolutions for instance segmentation," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 282–298.

[34] B. Cheng, A. Schwing, and A. Kirillov, "Per-pixel classification is not all you need for semantic segmentation," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 17864–17875.

[35] B. Cheng, I. Misra, A. G. Schwing, A. Kirillov, and R. Girdhar, "Masked-attention mask transformer for universal image segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 1280–1289.

[36] S. Wei, S. Ji, and M. Lu, "Toward automatic building footprint delineation from aerial images using CNN and regularization," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 3, pp. 2178–2189, Mar. 2020.

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[37] S. Wei, T. Zhang, S. Ji, M. Luo, and J. Gong, "BuildMapper: A fully learnable framework for vectorized building contour extraction," *ISPRS J. Photogramm. Remote Sens.*, vol. 197, pp. 87–104, Mar. 2023.

[38] L. Mou and X. X. Zhu, "Vehicle instance segmentation from aerial image and video using a multitask learning residual fully convolutional network," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 11, pp. 6699–6711, Nov. 2018.

[39] T. Zhang et al., "Semantic attention and scale complementary network for instance segmentation in remote sensing images," *IEEE Trans. Cybern.*, vol. 52, no. 10, pp. 10999–11013, Oct. 2022.

[40] S. Chen, Z. Zhang, R. Zhong, L. Zhang, H. Ma, and L. Liu, "A dense feature pyramid network-based deep learning model for road marking instance segmentation using MLS point clouds," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 1, pp. 784–800, Jan. 2021.

[41] Y. Gong, F. Zhang, X. Jia, Z. Mao, X. Huang, and D. Li, "Instance segmentation in very high resolution remote sensing imagery based on hard-to-segment instance learning and boundary shape analysis," *Remote Sens.*, vol. 14, no. 1, p. 23, Dec. 2021.

[42] W. Ye, W. Zhang, W. Lei, W. Zhang, X. Chen, and Y. Wang, "Remote sensing image instance segmentation network with transformer and multi-scale feature representation," *Expert Syst. Appl.*, vol. 234, Dec. 2023, Art. no. 121007.

[43] F. Yang et al., "Accurate instance segmentation for remote sensing images via adaptive and dynamic feature learning," *Remote Sens.*, vol. 13, no. 23, p. 4774, Nov. 2021.

[44] R. Bommasani et al., "On the opportunities and risks of foundation models," 2021, *arXiv:2108.07258*.

[45] J. Li, J. Wang, Z. Zhang, and H. Zhao, "Self-prompting large language models for zero-shot open-domain QA," 2022, *arXiv:2212.08635*.

[46] T. B. Brown et al., "Language models are few-shot learners," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 1877–1901.

[47] T. Chen et al., "SAM-adapter: Adapting segment anything in underperformed scenes," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2023, pp. 3367–3375.

[48] C. Li, P. Khanduri, Y. Qiang, R. Ibn Sultan, I. Chetty, and D. Zhu, "AutoProSAM: Automated prompting SAM for 3D multi-organ segmentation," 2023, *arXiv:2308.14936*.

[49] K. Zhang and D. Liu, "Customized segment anything model for medical image segmentation," 2023, *arXiv:2304.13785*.

[50] W. Liu, X. Shen, C.-M. Pun, and X. Cun, "Explicit visual prompting for low-level structure segmentations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2023, pp. 19434–19445.

[51] X. Lin, Y. Xiang, L. Zhang, X. Yang, Z. Yan, and L. Yu, "SAMUS: Adapting segment anything model for clinically-friendly and generalizable ultrasound image segmentation," 2023, *arXiv:2309.06824*.

[52] E. Hu et al., "LoRA: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learn. Represent.*, 2022.

[53] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," in *Proc. Adv. Neural Inf. Process. Syst.*, 2023, pp. 10088–10115.

[54] T. Ren et al., "Grounded SAM: Assembling open-world models for diverse visual tasks," 2024, *arXiv:2401.14159*.

[55] S. Liu et al., "Grounding DINO: Marrying DINO with grounded pre-training for open-set object detection," 2023, *arXiv:2303.05499*.

[56] R. Zhang et al., "Personalize segment anything model with one shot," 2023, *arXiv:2305.03048*.

[57] L. P. Osco et al., "The segment anything model (SAM) for remote sensing applications: From zero to one shot," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 124, Nov. 2023, Art. no. 103540.

[58] R. Ranftl, A. Bochkovskiy, and V. Koltun, "Vision transformers for dense prediction," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 12159–12168.

[59] X. Zhu, W. Su, L. Lu, B. Li, X. Wang, and J. Dai, "Deformable DETR: Deformable transformers for end-to-end object detection," 2020, *arXiv:2010.04159*.

[60] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 765–781.

[61] H. W. Kuhn, "The Hungarian method for the assignment problem," in *50 Years of Integer Programming 1958-2008: From the Early Years to the State-of-the-Art*. Berlin, Germany: Springer, 2010, pp. 29–47.

[62] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[63] G. Cheng, J. Han, P. Zhou, and L. Guo, "Multi-class geospatial object detection and geographic image classification based on collection of part detectors," *ISPRS J. Photogramm. Remote Sens.*, vol. 98, pp. 119–132, Dec. 2014.

[64] M. Gerke, F. Rottensteiner, J. Wegner, and G. Sohn, "ISPRS semantic labeling contest," ISPRS Working Group III/4, Hannover, Germany, Rep., Aug. 2014, doi: 10.13140/2.1.3570.9445. [Online]. Available: https://www.isprs.org/education/benchmarks/UrbanSemLab/default.aspx

[65] H. Su, S. Wei, M. Yan, C. Wang, J. Shi, and X. Zhang, "Object detection and instance segmentation in remote sensing imagery based on precise mask R-CNN," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Jul. 2019, pp. 1454–1457.

[66] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Represent.*, 2019.

[67] C. Zhang et al., "Faster segment anything: Towards lightweight SAM for mobile applications," 2023, *arXiv:2306.14289*.

[68] X. Zhao et al., "Fast segment anything," 2023, *arXiv:2306.12156*.



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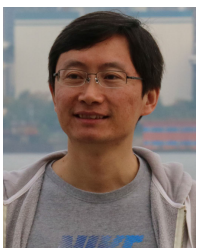
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